MIL-E-1/1069F
19 February 1981
SUPERSEDING
MIL-E-1/1069E
26 October 1970

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5702WB

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, subminiature, sharp cutoff

Outline	---	3-6 (EIA)
Base	---	Pinch press (7 leads)
Envelope	---	T3
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	a	g2	h	h	g3	k	g1

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Ehk	Rk	Rg1	Ik	Pp	Pg2	TE	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	v	Ohms	Meg	mAdc	W	W	°C	ft
Maximum:	6.9	165	---	155	0	200	---	1.2	16.5	1.3	0.5	220	(see note 4)
Minimum:	5.7	---	-55	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	120	0	120	0	0	200	0	---	---	---	---	---

GENERAL:

Qualification - Required

Reliable tube

5702wb

FSC 5960

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 1								
1256	Electrode current (1) (anode)			0.65	II	Ib	5.5	9.5	mAdc
1256	Electrode current (2) (anode)		Ec1 = -9.0 Vdc; Rk = 0	0.65	II	Ib	---	50	µAdc
1256	Electrode current (screen grid)			0.65	II	Ic2	1.7	3.5	mAdc
1266	Total grid current	1	Rg1 = 1.0 Meg	0.65	II	Ic1	0	-0.1	µAdc
1301	Heater current			0.65	II	If	190	210	mA
1306	Transconductance (1)			0.65	II	Sm	4,200	5,800	µmhos
1336	Heater-cathode leakage			0.65	II	Ihk	---	5.0	µAdc
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes			2.5	S3	R	250	---	Meg
1031	High-frequency vibration		Rp = 10,000 ohms	2.5	I	Ep	---	50	mVac
1246	Audio frequency noise		Esig = 70 mVac; Ec2 = 25 Vdc; Rg1 = 0.1 Meg; Rg2 = 1,000 ohms; Rp = 0.2 Meg; Rk = 4,000 ohms	2.5	I	EB	---	17	vu
1266	Grid emission	3	Ef = 7.5 V; Rg1 = 1.0 Meg; Ec1 = -10 Vdc	---	---	Ic1	0	-0.5	µAdc
1306	Transconductance (2)		Ef = 5.7 V	2.5	I	ΔSm Ef	---	5.0	%
1311	Anode resistance			---	---	rp	0.15	---	Meg
1331	Direct-interelectrode capacitance		0.405-inch diameter shield	6.5	Code E	Cg1p Cin Cout	--- 4.1 2.9	0.03 5.5 4.1	pF pF pF
1256	Electrode current (1) (anode) (tube operation time)		t = 20 seconds	4.0	S3	Ib	90	110	%
1031	Sweep-frequency vibration		Rp = 10,000 ohms, F = 30 to 1,000 Hz; 15 G; t = 3 minutes; positions X and Y only	4.0	S3	Ep	---	240	mv p to p

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 2 - continued								
1116	Lead fatigue			6.5	Code F	---	---	---	---
2126	Envelope strain			6.5	I	---	---	---	---
1041	Shock	5	450 G, Ehk = 100 Vdc	6.5	See note 2	---	---	---	---
1031	Vibration fatigue (1)			6.5	See note 2	---	---	---	---
1031	Vibration fatigue (2)		10 G, t = 6 hours	6.5	See note 2	---	---	---	---
---	Shock and vibration-fatigue (1) and (2)-test end points:								
1031	High-frequency vibration			---	---	Ep	---	75	mVac
1336	Heater-cathode leakage			---	---	Ihk	---	10	µAdc
1306	Change in trans-conductance (1) of individual tubes			---	---	ΔSm t	---	15	%
1266	Total grid current			---	---	Ic1	0	-1.0	µAdc
1119	Glass-to-lead seal strain			6.5	Code F	---	---	---	---
1105	Permanence of marking			---	---	---	---	---	---

Symbol	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.5 V; Ehk = 140 Vac, Eb = Ec1 = Ec2 = Ec3 = 0	---	---	---	---
---	Heater-cycling life-test end point:						
1336	Heater-cathode leakage			Ihk	---	20	µAdc
1516	Stability life		Ehk = 200 Vdc; Rgl = 1.0 Meg, TA = room	---	---	---	---
---	Stability life-test end point:						
1306	Change in transconductance (1) of individual tubes			ΔSm t	---	10	%
1501	Intermittent life	6	Ehk = 200 Vdc; Rgl = 1.0 Meg, TA = room; TE = 200°C (min); Group E	---	---	---	---

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3 - Continued						
---	Intermittent life-test end points (500 hours):						
1266	Total grid current			Ic1	0	-0.5	μAdc
1301	Heater current			If	187	217	mA
1306	Change in transconductance (1) of individual tubes			ΔSmt	---	20	%
1306	Transconductance (2)			ΔSmEf	---	15	%
1336	Heater-cathode leakage			Ihk	---	10	μAdc
1211	Insulation of electrodes			R	50	---	Meg
1306	Transconductance (1), average change			AvgΔSmt	---	15	%
---	Intermittent life-test end points (1,000 hours):						
1266	Total grid current			Ic1	0	-1.0	μAdc
1301	Heater current			If	177	223	mA
1306	Change in transconductance (1) of individual tubes			ΔSmt	---	30	%
1336	Heater-cathode leakage			Ihk	---	15	μAdc

NOTES:

1. This test shall be performed at the conclusion of the holding period.
2. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot has passed. MIL-STD-105, sample size code letter E, shall apply.
3. Prior to this test, tubes shall be preheated 5 minutes at the conditions indicated below. The 3-minute test shall not be permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid emission test.

Ef	Eb	Ec1	Ec2	Ec3	Rk	Rgl
V	Vdc	Vdc	Vdc	Vdc	Ohms	Meg
7.5	120	0	120	0	200	1.0

4. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
5. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is used.
6. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.
7. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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